



COMPUTER NETWORKS

23CSE302

LAB REPORT

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INDEX

S.no	Topic	Page.no	Signature
1	Commands regarding CISCO		
2	Topologies		
3	LAN Implementation		
4	MAN Implementation		
5	WAN Implementation		
6	Virtual LAN Implementation		
7	HTTP and HTTPS Implementation		
8	FTP Implementation		
9	DNS Implementation		
10	TCP and UDP Implementation		

LAB – 1

Aim:

Learn basic Cisco Networking commands like enable, hostname, interface, IP address, ping and HTTP commands like GET, POST, PUT and DELETE by practicing Cisco Packet Tracer

Apparatus:

1. Cisco Packet Tracer software
2. Router
3. Switch
4. PC
5. Server
6. Copper Straight-Through cables

Procedure:

Basic commands which I chosen

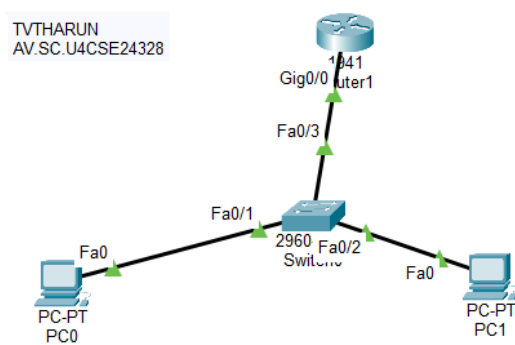
No.	Command	Definition
1	enable	Enters privileged EXEC mode, where you can type advanced router commands.
2	configure terminal	Enters global configuration mode on the router to make configurations.
3	hostname R1	Renaming the router to R1.
4	interface gigabit Ethernet 0/0	Configures Gigabit Ethernet 0/0 interface.
5	IP address 192.168.1.1 255.255.255.0	Assigns an IP address and subnet mask to the selected interface.
6	no shutdown	Activates/enables the selected network interface.
7	show IP interface brief	Displays a summary of interfaces, IP addresses, and their status.
8	ping 192.168.1.2	Tests the network connection between the router and the IP address you specify.
9	show running-config	Displays the current configuration of the router.

No.	Command	Definition
10	exit	Exits the current configuration mode and returns to the previous mode.

1. Create the topology

1. Open **Cisco Packet Tracer**.
2. Add **1 Router**.
3. Add **2 PCs**.
4. Add a **Switch**.
5. Connect:
 - PC0 → Switch
 - PC1 → Switch
 - Switch → Router

Use **Copper Straight-Through** cables.



2. Configure the Router

Click **Router** → **CLI** and enter:

```
Router> enable
```

```
Router# configure terminal
```

```
Router(config)# hostname R1
```

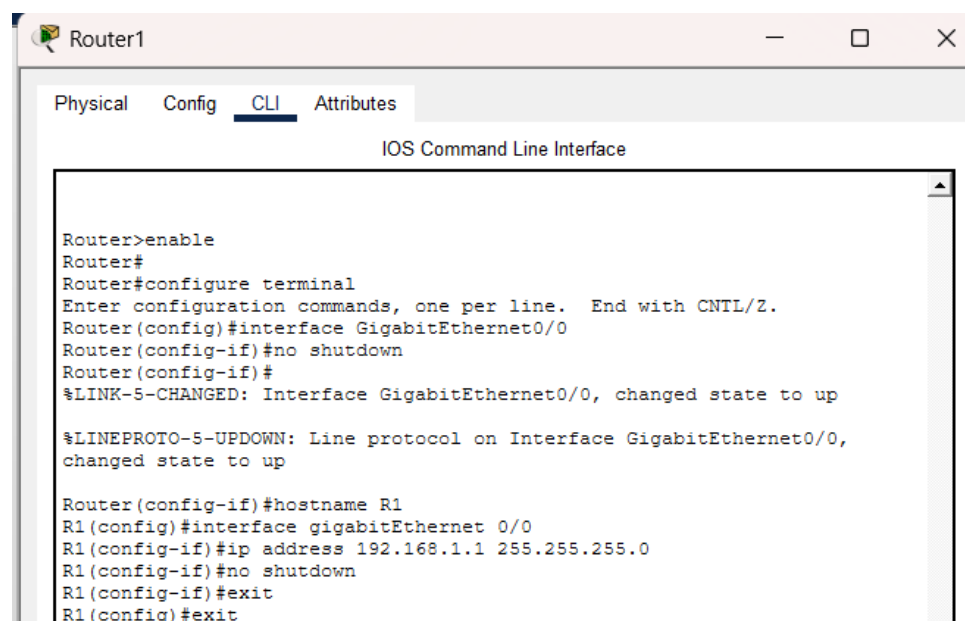
```
R1(config)# interface gigabit Ethernet 0/0
```

```
R1(config-if)# IP address 192.168.1.1 255.255.255.0
```

```
R1(config-if)# no shutdown
```

```
R1(config-if)# exit
```

```
R1(config)# exit
```



3. Check the interface

R1# show IP interface brief

You should see the interface as **up/up**.

```
R1#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  192.168.1.1    YES manual up          up
GigabitEthernet0/1  unassigned     YES unset  administratively down down
Vlan1           unassigned     YES unset  administratively down down
```

4. Configure PC0

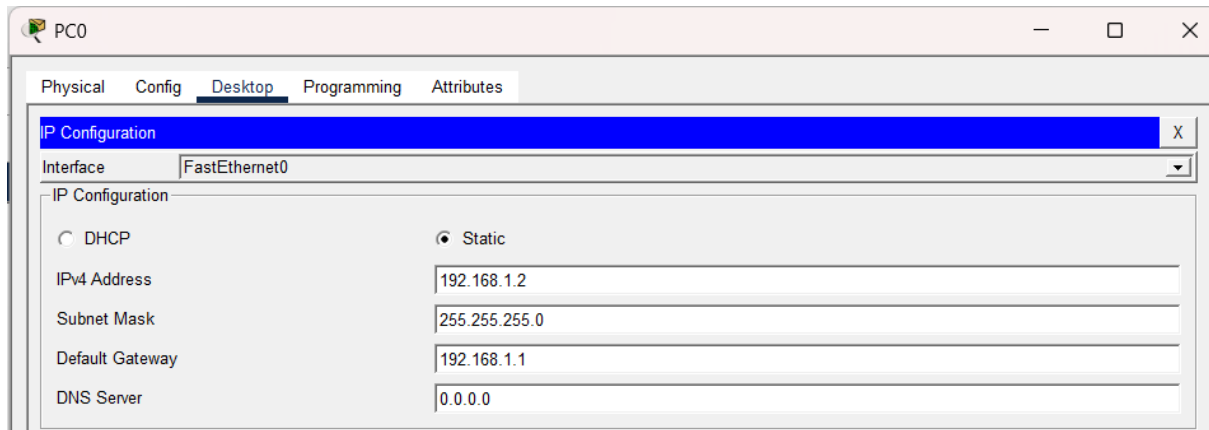
Click **PC0** → **Desktop** → **IP Configuration**.

Enter:

IP Address: 192.168.1.2

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1



5. Configure PC1

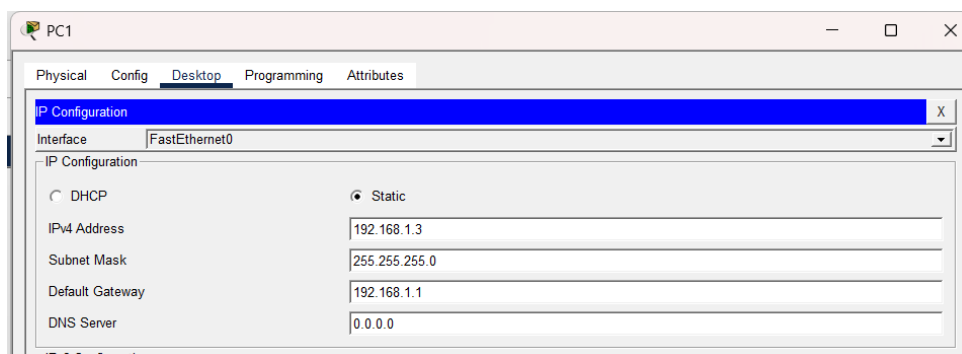
Click **PC1** → **Desktop** → **IP Configuration**.

Enter:

IP Address: 192.168.1.3

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1



6. Test connectivity

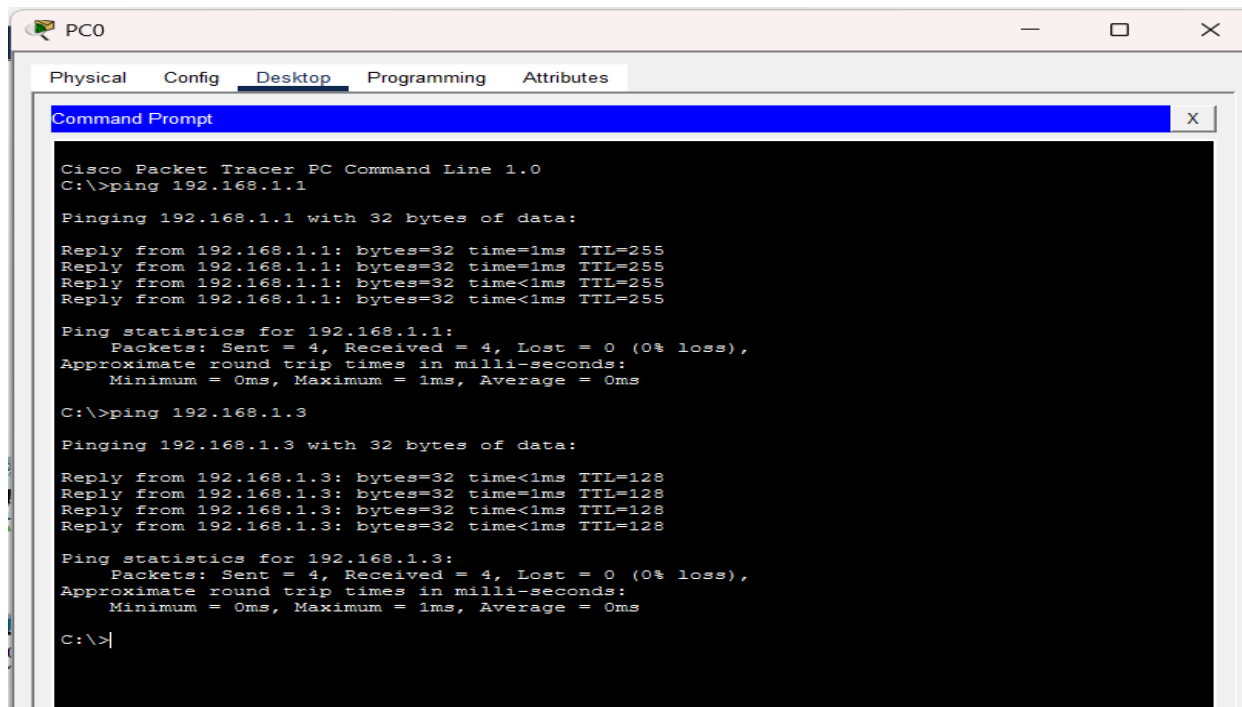
From **PC0** → **Desktop** → **Command Prompt**, enter:

ping 192.168.1.1

Then:

ping 192.168.1.3

If you receive **Reply from...**, the connection is working.



7. View the router configuration

On the router CLI:

R1# show running-config

This displays the configuration you have created.

```

R1#show running-config
Building configuration...

Current configuration : 624 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname R1
!
!
!
!
!
!
no ip cef
no ipv6 cef
!
!
--More--

```

Result:

The basic Cisco router configuration commands were successfully implemented in Cisco Packet Tracer. The router was configured with a hostname and IP address, the Gigabit Ethernet interface was enabled, and the interface status was verified. Connectivity was successfully tested using the ping command.

LAB – 2

Aim:

Discover and utilize different network topologies using Cisco Packet Tracer.

To study the link and communication between devices in different topologies.

Apparatus:

1. Cisco Packet Tracer
2. PCs / End Devices
3. Switches
4. Routers
5. Ethernet cables
6. IP addresses for network configuration
7. Command Prompt / CLI

Procedure:

1.Bus Topology

Step 1: Build the workspace

1. Start Cisco Packet Tracer.
2. Make a new workspace.

Step 2: Add devices

Add the following devices:

- PC0
- PC1
- PC2
- PC3
- Switch0
- Switch1
- Switch2
- Switch3

Step 3: Connect the switches

Connect the switches in a straight line:

Switch0 → Switch16 → Switch17 → Switch18

Use Copper Straight-Through cables.

Step 4: Connect PCs to switches

PC	Connected Switch
PC0	Switch0
PC1	Switch16
PC2	Switch17
PC3	Switch18

Connect each PC's Fast Ethernet port to the switch using a Copper Straight-Through cable.

Step 5: Configure IP addresses

PC0

PC0 → Desktop → IP Configuration → Static → IP Address: 10.10.10.1 → Subnet Mask: 255.0.0.0 → Default Gateway: (leave empty) → Close window

PC1

PC1 → Desktop → IP Configuration → Static → IP Address: 10.10.10.2 → Subnet Mask: 255.0.0.0 → Default Gateway: (leave empty) → Close window

PC2

PC2 → Desktop → IP Configuration → Static → IP Address: 10.10.10.3 → Subnet Mask: 255.0.0.0 → Default Gateway: (leave empty) → Close window

PC3

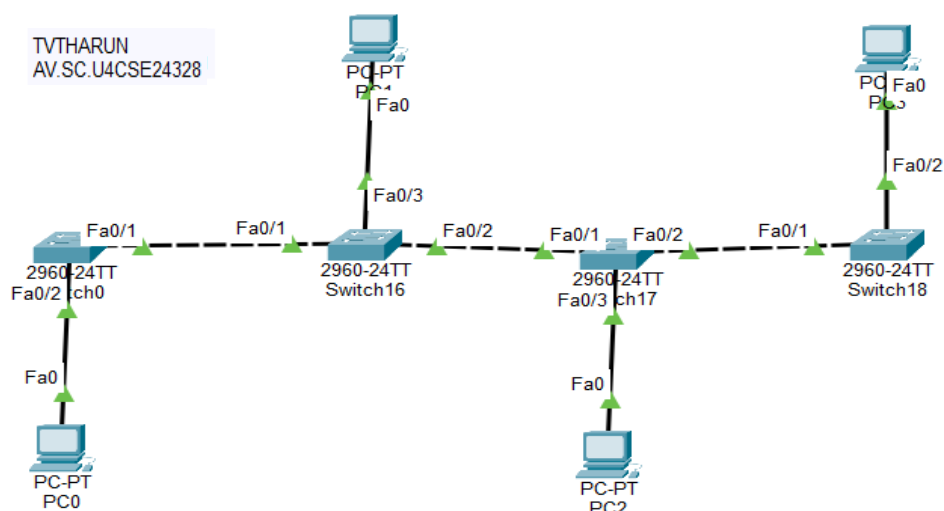
PC3 → Desktop → IP Configuration → Static → IP Address: 10.10.10.4 → Subnet Mask: 255.0.0.0 → Default Gateway: (leave empty) → Close window

Step 6: Test connectivity

1. Open PC0 → Desktop → Command Prompt.
2. Enter:

ping 10.10.10.4
3. You should receive Reply from 192.168.10.4.

OUTPUT:



Test cases:

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC1	ICMP		0.000	N	0	(edit)	(delete)

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC0	ICMP
	0.001	PC0	Switch0	ICMP
	0.002	Switch0	Switch16	ICMP
	0.003	Switch16	PC1	ICMP
	0.004	PC1	Switch16	ICMP
	0.005	Switch16	Switch0	ICMP
	0.006	Switch0	PC0	ICMP
	1.994	--	Switch18	STP

Failed cases:

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Failed	PC9	PC11	ICMP		0.000	N	0	(edit)	(delete)

1. If IP addresses of the end devices are not given.
2. Connections are not provided.
3. if copper cross over is provided instead of copper straight-through and vice versa

2. Ring Topology

1. Open Cisco Packet Tracer and create a new workspace.
2. Add 4 PCs:
 - PC8
 - PC9
 - PC10
 - PC11
3. Add 4 switches:
 - Switch0
 - Switch1
 - Switch2
 - Switch3Use 2960-24TT switches.
4. Arrange the switches in a closed-loop/line-like bus arrangement as shown in the figure.
5. Connect the switches using Copper Straight-Through cables:
 - Switch0 → Switch2
 - Switch2 → Switch3
 - Switch3 → Switch4
 - Switch4 → Switch0
6. Connect the PCs to the switches:
 - PC8 → Switch0
 - PC9 → Switch2
 - PC10 → Switch3
 - PC11 → Switch4

7. Use FastEthernet0 on each PC and connect it to an available Fast Ethernet port on the corresponding switch.

8. Configure the IP addresses:

PC8:

PC8 → Desktop → IP Configuration → Static → IP Address: 11.11.11.1 → Subnet Mask: 255.0.0.0 → Default Gateway: (leave empty) → Close window

PC9:

PC9 → Desktop → IP Configuration → Static → IP Address: 11.11.11.2 → Subnet Mask: 255.0.0.0 → Default Gateway: (leave empty) → Close window

PC10:

PC10 → Desktop → IP Configuration → Static → IP Address: 11.11.11.3 → Subnet Mask: 255.0.0.0 → Default Gateway: (leave empty) → Close window

PC11:

PC11 → Desktop → IP Configuration → Static → IP Address: 11.11.11.4 → Subnet Mask: 255.0.0.0 → Default Gateway: (leave empty) → Close window

9. Wait until all connected links become green.

10. Test connectivity from PC8 → Desktop → Command Prompt using:

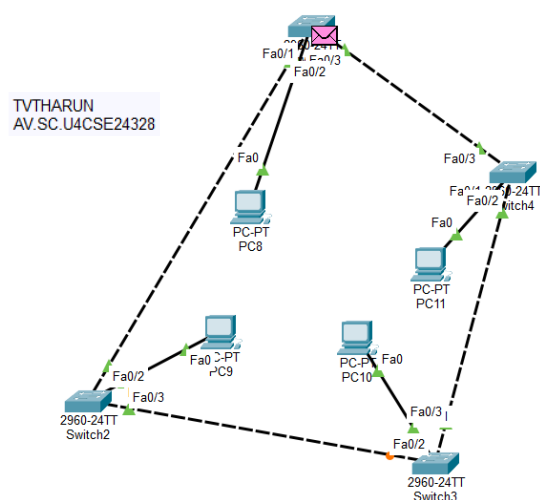
ping 11.11.11.2

ping 11.11.11.3

ping 11.11.11.4

11. If Reply from... is received, the connectivity is successful.

OUTPUT:



Test cases:

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC9	PC11	ICMP		0.000	N	0	(edit)	(delete)

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC9	ICMP
	0.001	PC9	Switch2	ICMP
	0.002	Switch2	Switch1	ICMP
	0.003	Switch1	Switch4	ICMP
	0.004	Switch4	PC11	ICMP
	0.005	PC11	Switch4	ICMP
	0.006	Switch4	Switch1	ICMP
	0.007	Switch1	Switch2	ICMP
	0.008	Switch2	PC9	ICMP
	1.840	--	Switch1	STP

Failed Case:

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Failed	PC9	PC11	ICMP		0.000	N	0	(edit)	(delete)

1. If IP addresses of the end devices are not given.
2. If the connections are not provided.
3. if copper straight-through is given instead of copper cross-over and vice versa.

3.Star Toplogy

1. Open Cisco Packet Tracer and create a new workspace.
2. Add 1 switch:
 - Cisco 2960-24TT Switch
3. Add 4 PCs:
 - PC12
 - PC13
 - PC14
 - PC15
4. Place the switch at the center and arrange the four PCs around it as shown in the figure.
5. Connect each PC directly to the switch using Copper Straight-Through cables:

- PC13 → Switch Fa0/1
- PC12 → Switch Fa0/2
- PC15 → Switch Fa0/3
- PC14 → Switch Fa0/4

6. Configure the IP addresses of the PCs:

PC 12:

PC 12 → Desktop → IP Configuration → Static → IP Address: 12.12.12.1 → Subnet Mask: 255.0.0.0 → Default Gateway: (leave empty) → Close window

PC 13:

PC 13 → Desktop → IP Configuration → Static → IP Address: 12.12.12.2 → Subnet Mask: 255.0.0.0 → Default Gateway: (leave empty) → Close window

PC 14:

PC 14 → Desktop → IP Configuration → Static → IP Address: 12.12.12.3 → Subnet Mask: 255.0.0.0 → Default Gateway: (leave empty) → Close window

PC 15:

PC 15 → Desktop → IP Configuration → Static → IP Address: 12.12.12.4 → Subnet Mask: 255.0.0.0 → Default Gateway: (leave empty) → Close window

7. Wait until all the links between the PCs and switch become green.

8. Test connectivity from PC12 using:

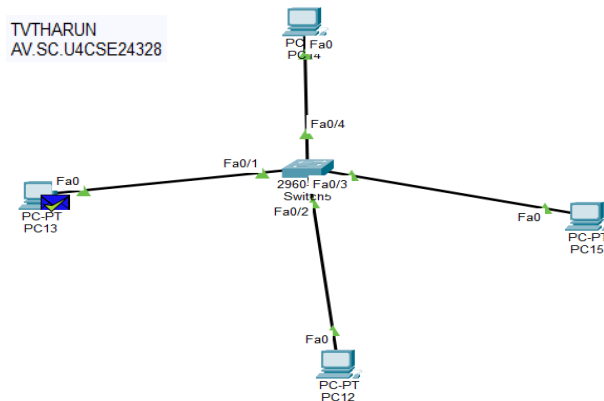
ping 12.12.12.2

ping 12.12.12.3

ping 12.12.12.4

9.If replies are received, the connectivity between all PCs is successfully established.

OUTPUT:



Test cases:

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC13	PC12	ICMP		0.000	N	0	(edit)	(delete)

Simulation Panel				
Event List				
Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC13	ICMP
	0.001	PC13	Switch5	ICMP
	0.002	Switch5	PC12	ICMP
	0.003	PC12	Switch5	ICMP
	0.004	Switch5	PC13	ICMP

Failed Case:

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Failed	PC13	PC12	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC13	PC15	ICMP		0.000	N	1	(edit)	(delete)

1. Here one PC and switch connection was removed.
2. When I tried to transfer packet, it failed when I chose connectionless PC.
3. It shows successful when I choose connected PC.
4. This is an advantage of the star topology 11.

4.Mesh Topology

1. Open Cisco Packet Tracer and create a new workspace.
2. Add 4 switches:

- Switch5
 - Switch6
 - Switch8
 - Switch9
- Use 2960-24TT switches.

3. Add 4 PCs:

- PC16
- PC17
- PC18
- PC19

4. Arrange the four switches in a square as shown in the figure.

5. Connect the switches to form the mesh:

- Switch5 → Switch6
- Switch6 → Switch9
- Switch9 → Switch8
- Switch8 → Switch5
- Switch5 → Switch9
- Switch6 → Switch8

6. Use Copper Straight-Through cables for all switch-to-switch connections.

7. Connect each PC to its corresponding switch:

- PC16 → Switch5
- PC17 → Switch6
- PC18 → Switch8
- PC19 → Switch9

8. Configure the IP addresses:

9. PC 16:

PC 16 → Desktop → IP Configuration → Static → IP Address: 13.13.13.1 → Subnet Mask: 255.0.0.0 → Default Gateway: (leave empty) → Close window

PC 17:

PC 17 → Desktop → IP Configuration → Static → IP Address: 13.13.13.2 → Subnet Mask: 255.0.0.0 → Default Gateway: (leave empty) → Close window

PC 18:

PC 18 → Desktop → IP Configuration → Static → IP Address: 13.13.13.3 → Subnet Mask: 255.0.0.0 → Default Gateway: (leave empty) → Close window

PC 19:

PC 19 → Desktop → IP Configuration → Static → IP Address: 13.13.13.4 → Subnet Mask: 255.0.0.0 → Default Gateway: (leave empty) → Close window

10. Wait until the links become green.

11. Test connectivity from PC16 → Desktop → Command Prompt:

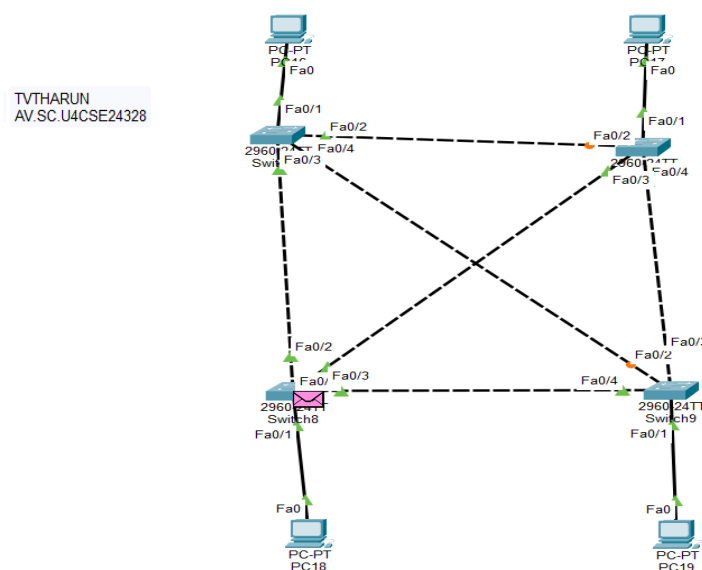
ping 192.168.10.2

ping 192.168.10.3

ping 192.168.10.4

12. If replies are received, connectivity between the PCs is successfully established.

OUTPUT:



Simulation Panel				
Event List				
Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC18	ICMP
	0.001	PC18	Switch8	ICMP
	0.002	Switch8	Switch7	ICMP
	0.003	Switch7	PC17	ICMP
	0.004	PC17	Switch7	ICMP
	0.005	Switch7	Switch8	ICMP
	0.006	Switch8	PC18	ICMP
	0.939	--	Switch8	STP

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC18	PC17	ICMP		0.000	N	0	(edit)	(delete)

Failed Case:

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC18	PC16	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC18	PC17	ICMP		0.000	N	1	(edit)	(delete)

1. After disconnecting the two routers, the packet transfer was successful.
2. Due to the mesh connection between routers, each router was connected to each other router in the topology.
3. Packet loss is very rare, if it is lost then the IP addresses are incorrectly allotted.

5.Tree Topology

1. Open Cisco Packet Tracer and create a new workspace.
2. Add 7 switches:
 - 1 main switch at the top
 - 2 intermediate switches
 - 4 lower-level switches
 Use 2960-24TT switches.
3. Add 4 PCs:
 - PC20
 - PC21
 - PC22
 - PC23

4. Arrange the devices in a hierarchical tree structure as shown in the figure.
5. Connect the main switch to the two intermediate switches:
 - Main Switch → Left Intermediate Switch
 - Main Switch → Right Intermediate Switch
6. Connect the left intermediate switch to two lower-level switches:
 - Left Intermediate Switch → Switch12
 - Left Intermediate Switch → Switch13
7. Connect the right intermediate switch to two lower-level switches:
 - Right Intermediate Switch → Switch14
 - Right Intermediate Switch → Switch15
8. Connect the PCs to the lower-level switches:
 - PC20 → Switch12
 - PC21 → Switch13
 - PC22 → Switch14
 - PC23 → Switch15
9. Use Copper Straight-Through cables for all connections.
10. Configure the IP addresses:

PC 20:
PC 20 → Desktop → IP Configuration → Static → IP Address: 14.14.14.1 → Subnet Mask: 255.0.0.0 → Default Gateway: (leave empty) → Close window

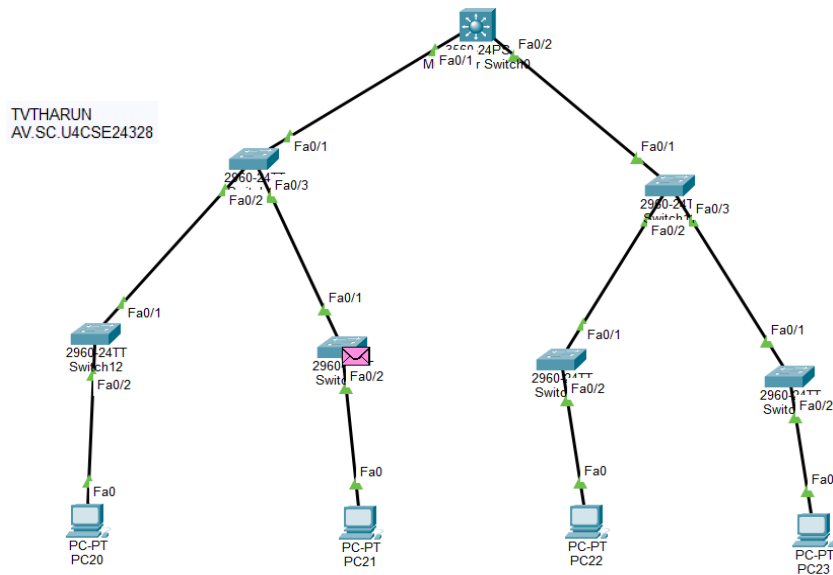
PC 21:
PC 21 → Desktop → IP Configuration → Static → IP Address: 14.14.14.2 → Subnet Mask: 255.0.0.0 → Default Gateway: (leave empty) → Close window

PC 22:
PC 22 → Desktop → IP Configuration → Static → IP Address: 14.14.14.3 → Subnet Mask: 255.0.0.0 → Default Gateway: (leave empty) → Close window

PC 23:
PC 23 → Desktop → IP Configuration → Static → IP Address: 14.14.14.4 → Subnet Mask: 255.0.0.0 → Default Gateway: (leave empty) → Close window
11. Wait until all the connections become green.
12. Test connectivity from PC20 → Desktop → Command Prompt:
 - ping 192.168.10.2
 - ping 192.168.10.3
 - ping 192.168.10.4

13. If replies are received, connectivity between the PCs is successfully established.

OUTPUT:



Vis.	Time(sec)	Last Device	At Device	Type
0.000	--	PC20	PC20	ICMP
0.001	PC20	Switch12	Switch12	ICMP
0.002	Switch12	Switch10	Switch10	ICMP
0.003	Switch10	Multilayer Switch0	Multilayer Switch0	ICMP
0.004	Multilayer Switch0	Switch11	Switch11	ICMP
0.005	Switch11	Switch14	Switch14	ICMP
0.006	Switch14	PC22	PC22	ICMP
0.007	PC22	Switch14	Switch14	ICMP
0.008	Switch14	Switch11	Switch11	ICMP
0.009	Switch11	Multilayer Switch0	Multilayer Switch0	ICMP
0.010	Multilayer Switch0	Switch10	Switch10	ICMP
0.011	Switch10	Switch12	Switch12	ICMP
0.012	Switch12	PC20	PC20	ICMP
1.945	--	Switch13	Switch13	STP

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC20	PC22	ICMP		0.000	N	0	(edit)	(delete)

Failed Case:

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Failed	PC20	PC22	ICMP		0.000	N	0	(edit)	(delete)
	Failed	PC21	PC22	ICMP		0.000	N	1	(edit)	(delete)

1. When I removed the parent node(router) the whole connection was dismissed.
2. When I tried to transfer the packet from Pc in one node to the PC in another node.
3. The packet is lost transfer is failed.

LAB – 3

Aim:

To implement a **Local Area Network (LAN)** using a switch and four PCs in Cisco Packet Tracer and verify connectivity between the devices using the ping command.

Apparatus:

1. Cisco Packet Tracer
2. 1 × Cisco 2960-24TT Switch
3. 4 × PCs
4. Copper Straight-Through cables
5. IP addresses
6. Command Prompt for testing connectivity

Procedure:

1. Open Cisco Packet Tracer and create a new workspace.
2. Add 1 switch:
 - Cisco 2960-24TT Switch
3. Add 4 PCs:
 - PC0
 - PC1
 - PC2
 - PC3
4. Place the switch at the center and the four PCs around it as shown in the figure.
5. Connect each PC to the switch using Copper Straight-Through cables:
 - PC0 → Switch0 Fa0/1
 - PC1 → Switch0 Fa0/2
 - PC2 → Switch0 Fa0/3
 - PC3 → Switch0 Fa0/4
6. Configure the IP addresses:
PC0:
PC0 → Desktop → IP Configuration → Static → IP Address: 10.11.12.1 → Subnet Mask: 255.0.0.0 → Default Gateway: (leave empty) → Close window
PC1:
PC1 → Desktop → IP Configuration → Static → IP Address: 10.11.12.2 → Subnet Mask: 255.0.0.0 → Default Gateway: (leave empty) → Close window

PC2:

PC2 → Desktop → IP Configuration → Static → IP Address: 10.11.12.3 → Subnet Mask: 255.0.0.0 → Default Gateway: (leave empty) → Close window

PC3:

PC3 → Desktop → IP Configuration → Static → IP Address: 10.11.12.4 → Subnet Mask: 255.0.0.0 → Default Gateway: (leave empty) → Close window

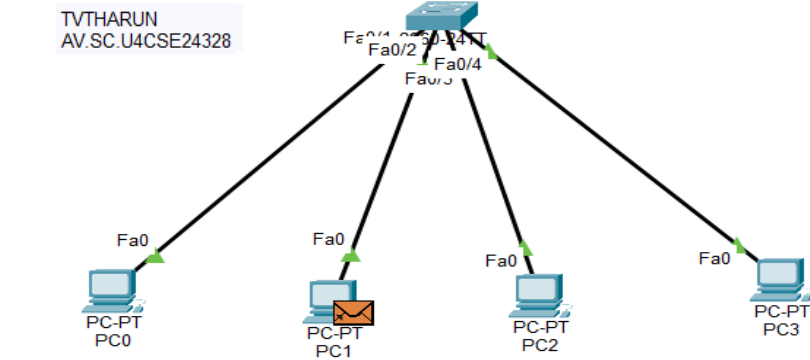
7. Wait until all connections become green.
8. Test the LAN connectivity from PC0 → Desktop → Command Prompt:

ping 10.11.12.2

ping 10.11.12.3

ping 10.11.12.4

11. If Reply from is displayed, the LAN connectivity is successful.

OUTPUT:


TVTHARUN
AV.SC.U4CSE24328

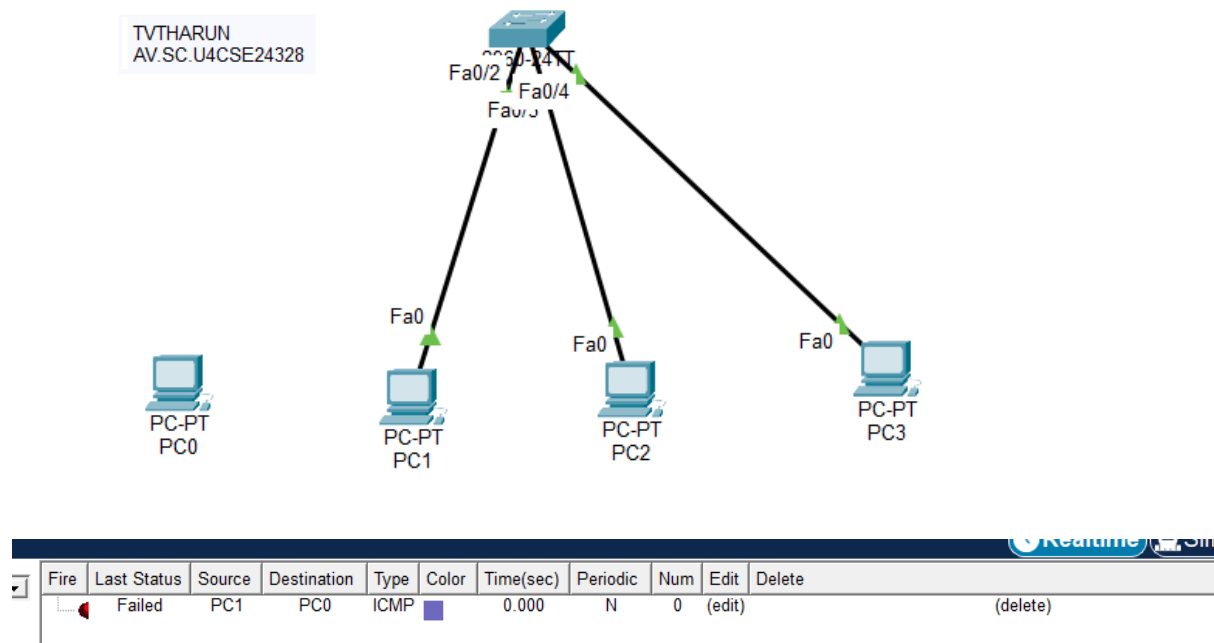
Simulation Panel

Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC1	ICMP
	0.001	PC1	Switch0	ICMP
	0.002	Switch0	PC2	ICMP
	0.003	PC2	Switch0	ICMP
	0.004	Switch0	PC1	ICMP

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC1	PC2	ICMP		0.000	N	0	(edit)	(delete)

Failed Case:



1. When I removed the connection between PC0 and PC1 it can't transfer the packet.
2. If the host number in IP address is not correctly assigned it show failed in status panel.

LAB – 4

Aim:

To implement a **Metropolitan Area Network (MAN)** using a router, switches, and PCs in Cisco Packet Tracer and verify communication between different LANs.

Apparatus:

- Cisco Packet Tracer
- 1 × Cisco ISR4331 Router
- 2 × Cisco 2960-24TT Switches
- 4 × PCs
- Copper Straight-Through cables
- IP addresses
- Command Prompt for connectivity testing

Procedure:

1. Open Cisco Packet Tracer and create a new workspace.
2. Add the following devices:

1 Router – Cisco ISR4331

2 Switches – Cisco 2960-24TT

4 PCs – PC4, PC5, PC6, PC7

3. Arrange the devices as shown in the figure.

Connect the router to the switches using Copper Straight-Through cables:

Router Gig0/0/0 → Switch1 Fa0/3

Router Gig0/0/1 → Switch2 Fa0/3

1. Connect the PCs to the switches:

PC4 → Switch1 Fa0/1

PC5 → Switch1 Fa0/2

PC6 → Switch2 Fa0/1

PC7 → Switch2 Fa0/2

- Configure the IP addresses of the PCs:

PC4:

PC4 → Desktop → IP Configuration → Static → IP Address: 10.1.1.1 → Subnet Mask: 255.0.0.0 → Default Gateway: 10.1.1.3 → Close window

PC5:

PC5 → Desktop → IP Configuration → Static → IP Address: 10.1.1.2 → Subnet Mask: 255.0.0.0 → Default Gateway: 10.1.1.3 → Close window

PC6:

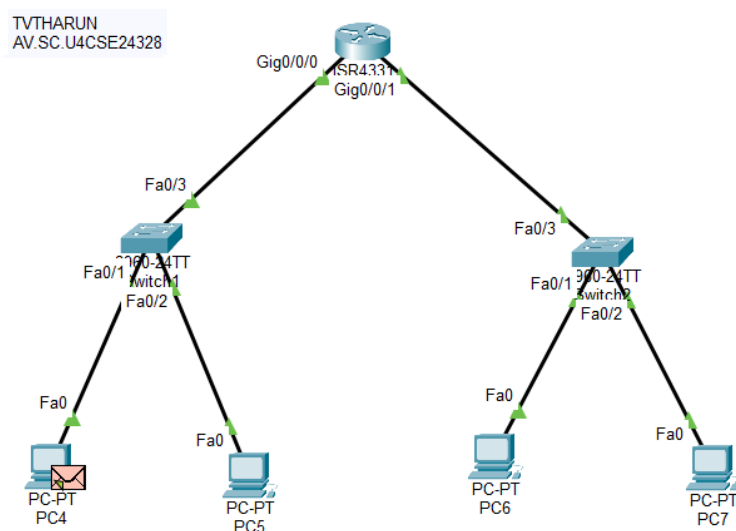
PC6 → Desktop → IP Configuration → Static → IP Address: 11.1.1.1 → Subnet Mask: 255.0.0.0 → Default Gateway: 11.1.1.3 → Close window

PC7:

PC7 → Desktop → IP Configuration → Static → IP Address: 11.1.1.2 → Subnet Mask: 255.0.0.0 → Default Gateway: 11.1.1.3 → Close window

- Check the router port status:
- Click **Router** → **Config** → **Gigabit Ethernet 0/0/0**
- Make sure **Port Status: On**
- Enter the IP address 10.1.1.3
- Click **Gigabit Ethernet 0/0/1**
- Make sure **Port Status: On**
- Enter the IP address 11.1.1.3
- Wait until all connected links become **green**.

OUTPUT:



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC4	PC7	ICMP		0.000	N	0	(edit)	(delete)

Simulation Panel

Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC4	ICMP
	0.001	PC4	Switch1	ICMP
	0.002	Switch1	Router0	ICMP
	0.003	Router0	Switch2	ICMP
	0.004	Switch2	PC7	ICMP
	0.005	PC7	Switch2	ICMP
	0.006	Switch2	Router0	ICMP
	0.007	Router0	Switch1	ICMP
	0.008	Switch1	PC4	ICMP

Failed Case:

Router0

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0/0

GigabitEthernet0/0/1

GigabitEthernet0/0/2

GigabitEthernet0/0/0

Port Status ☐ On

Link Speed ☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0090.0C51.7E01

IP Configuration

IPv4 Address 10.1.1.4

Subnet Mask 255.0.0.0

Tx Ring Limit 10

Make sure that the default gateway in PCs and IPv4 address in router matches. If there any mistake in addressing the IP address of router the packet will not transfer.

Check whether the port status is in ON.

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Failed	PC4	PC7	ICMP		0.000	N	0	(edit)	(delete)

LAB – 5

Aim:

To implement a Wide Area Network (WAN) using multiple routers, switches, and PCs in Cisco Packet Tracer and verify communication between different networks.

Apparatus:

- Cisco Packet Tracer
- 3 Routers
- 3 Cisco 2960-24TT Switches
- 6 PCs
- Copper Straight-Through cables
- Serial DCE/DTE cables
- IP addresses
- Command Prompt for connectivity testing

Procedure:

1. Open Cisco Packet Tracer and create a new workspace.
2. Add:
 - 3 Routers – Router6, Router7, Router8
 - 3 Switches – Switch6, Switch7, Switch8
 - 6 PCs – PC0 to PC5
3. Connect the devices as shown in the diagram:
 - Router6 Fa0/0 → Switch6 Fa0/3
 - Router7 Fa0/0 → Switch7 Fa0/3
 - Router8 Fa0/0 → Switch8 Fa0/3
 - Connect the PCs to their respective switches.
 - Connect Router6 Se2/0 → Router7 Se2/0.
 - Connect Router7 Se3/0 → Router8 Se2/0.
4. Assign IP addresses directly through the Router Config tab.

Router6

- Click Router6 → Config
- Select Fast Ethernet/Gigabit Ethernet 0/0
- IP Address: 192.168.1.1
- Subnet Mask: 255.255.255.0
- Turn Port Status: ON
- Select Serial 2/0
- IP Address: 10.0.0.1
- Subnet Mask: 255.255.255.252
- Turn Port Status: ON

Router7

- Click Router7 → Config
- Select Fast Ethernet/Gigabit Ethernet 0/0
- IP Address: 192.168.2.1
- Subnet Mask: 255.255.255.0
- Turn Port Status: ON
- Select Serial 2/0
- IP Address: 10.0.0.2
- Subnet Mask: 255.255.255.252
- Turn Port Status: ON
- Select Serial 3/0
- IP Address: 10.0.0.5
- Subnet Mask: 255.255.255.252
- Turn Port Status: ON

Router8

- Click Router8 → Config
- Select Fast Ethernet/Gigabit Ethernet 0/0
- IP Address: 192.168.3.1
- Subnet Mask: 255.255.255.0

- Turn Port Status: ON
- Select Serial 2/0
- IP Address: 10.0.0.6
- Subnet Mask: 255.255.255.252
- Turn Port Status: ON

5. Configure the PCs using PC → Desktop → IP Configuration → Static.

PC0:

PC0 → Desktop → IP Configuration → Static → IP Address: 192.168.1.2 → Subnet Mask: 255.255.255.0 → Default Gateway: 192.168.1.1 → Close window

PC1:

PC1 → Desktop → IP Configuration → Static → IP Address: 192.168.1.3 → Subnet Mask: 255.255.255.0 → Default Gateway: 192.168.1.1 → Close window

PC2:

PC2 → Desktop → IP Configuration → Static → IP Address: 192.168.2.2 → Subnet Mask: 255.255.255.0 → Default Gateway: 192.168.2.1 → Close window

PC3:

PC3 → Desktop → IP Configuration → Static → IP Address: 192.168.2.3 → Subnet Mask: 255.255.255.0 → Default Gateway: 192.168.2.1 → Close window

PC4:

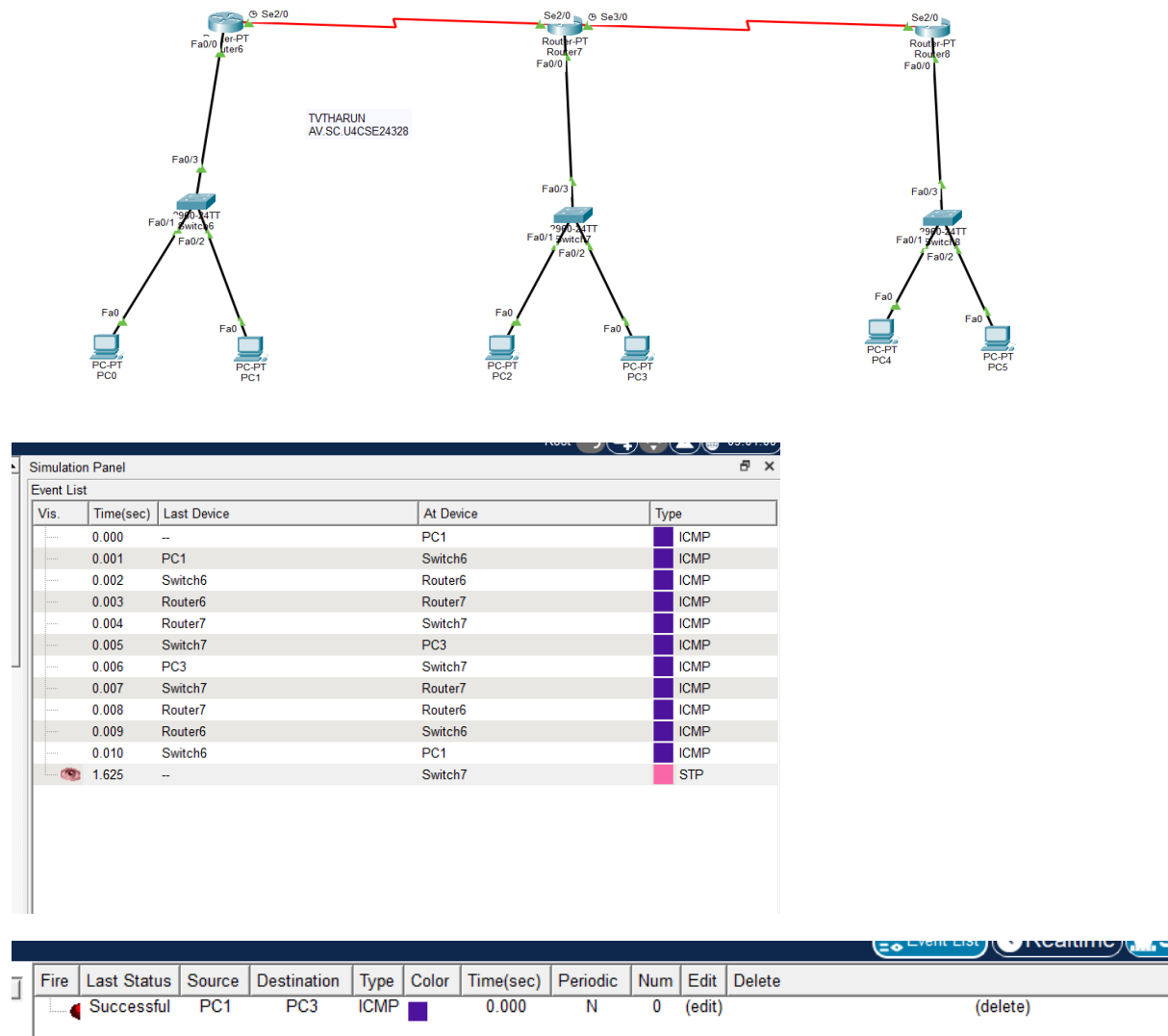
PC4 → Desktop → IP Configuration → Static → IP Address: 192.168.3.2 → Subnet Mask: 255.255.255.0 → Default Gateway: 192.168.3.1 → Close window

PC5:

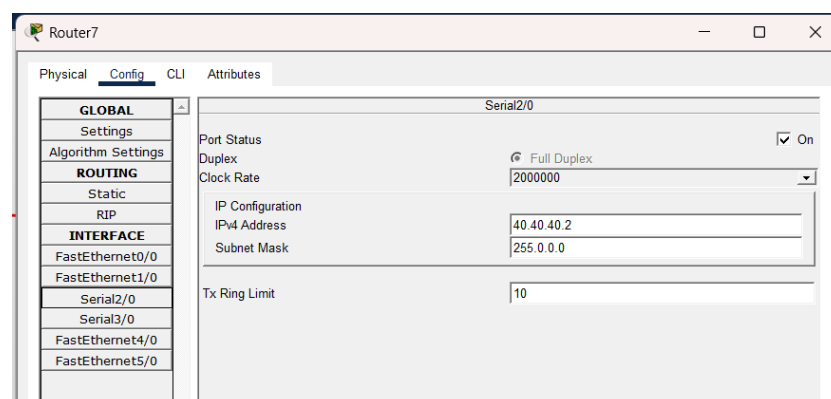
PC5 → Desktop → IP Configuration → Static → IP Address: 192.168.3.3 → Subnet Mask: 255.255.255.0 → Default Gateway: 192.168.3.1 → Close window

6. For the serial connections, configure the DCE side with an appropriate clock rate if required.
7. Check that all router ports are ON and the links become green.
8. Test connectivity using PC → Desktop → Command Prompt:
- ping 192.168.2.2
- ping 192.168.3.2
9. Successful replies indicate that communication between the networks has been established.

OUTPUT:



Failed Case:



If you make any mistake in configuring the IP addresses in the 3 routers the packet transfer will not work properly.

LAB – 6

Aim:

To set up VLANs on two switches in Cisco Packet Tracer, assign PCs to different VLANs, configure a trunk link, and check communication between devices in the same and different VLANs.

Apparatus:

- Cisco Packet Tracer
- 2 × Cisco 2960 switches
- 8 × PCs
- Copper Straight-Through cables
- Copper Cross-Over cable
- IP addresses
- Command Prompt for testing

Procedure:

1. Open Cisco Packet Tracer and create a new workspace.
2. Place 2 switches and 8 PCs in the workspace.
3. Connect the two switches using a Copper Cross-Over cable:
 - Switch3 Fa0/1 → Switch4 Fa0/1
4. Connect the PCs to the switches using Copper Straight-Through cables.
5. Assign IP addresses to the PCs.

CSE VLAN – VLAN 10

PC0:

PC0 → Desktop → IP Configuration → Static → IP Address: 192.168.10.1 → Subnet Mask: 255.255.255.0 → Default Gateway: (leave empty) → Close window

PC1:

PC1 → Desktop → IP Configuration → Static → IP Address: 192.168.10.2 → Subnet Mask: 255.255.255.0 → Default Gateway: (leave empty) → Close window

PC4:

PC4 → Desktop → IP Configuration → Static → IP Address: 192.168.10.3 → Subnet Mask: 255.255.255.0 → Default Gateway: (leave empty) → Close window

PC5:

PC5 → Desktop → IP Configuration → Static → IP Address: 192.168.10.4 → Subnet Mask: 255.255.255.0 → Default Gateway: (leave empty) → Close window

VLAN 20 (CAI VLAN):**PC2:**

PC2 → Desktop → IP Configuration → Static → IP Address: 192.168.20.1 → Subnet Mask: 255.255.255.0 → Default Gateway: (leave empty) → Close window

PC3:

PC3 → Desktop → IP Configuration → Static → IP Address: 192.168.20.2 → Subnet Mask: 255.255.255.0 → Default Gateway: (leave empty) → Close window

PC6:

PC6 → Desktop → IP Configuration → Static → IP Address: 192.168.20.3 → Subnet Mask: 255.255.255.0 → Default Gateway: (leave empty) → Close window

PC7:

PC7 → Desktop → IP Configuration → Static → IP Address: 192.168.20.4 → Subnet Mask: 255.255.255.0 → Default Gateway: (leave empty) → Close window

6. Configure Switch3 using the CLI.

enable

configure terminal

vlan 10

name CSE

exit

vlan 20

name CAI

exit

interface range fa0/2-3

switchport mode access

switchport access vlan 10

exit

interface range fa0/4-5

switchport mode access

switchport access vlan 20

exit

interface fa0/1

switchport mode trunk

exit

7. Configure Switch4 similarly.

enable

configure terminal

vlan 10

name CSE

exit

vlan 20

name CAI

exit

interface range fa0/2-3

switchport mode access

switchport access vlan 10

exit

interface range fa0/4-5

switchport mode access

switchport access vlan 20

exit

interface fa0/1

switchport mode trunk

exit

8. Verify the VLAN configuration using:

show vlan brief

9. Verify the trunk connection using:

show interfaces trunk

10. Test communication within the same VLAN.

From PC0, open Desktop → Command Prompt and enter:

ping 192.168.10.3

The ping should be successful because PC0 and PC4 belong to VLAN 10 (CSE).

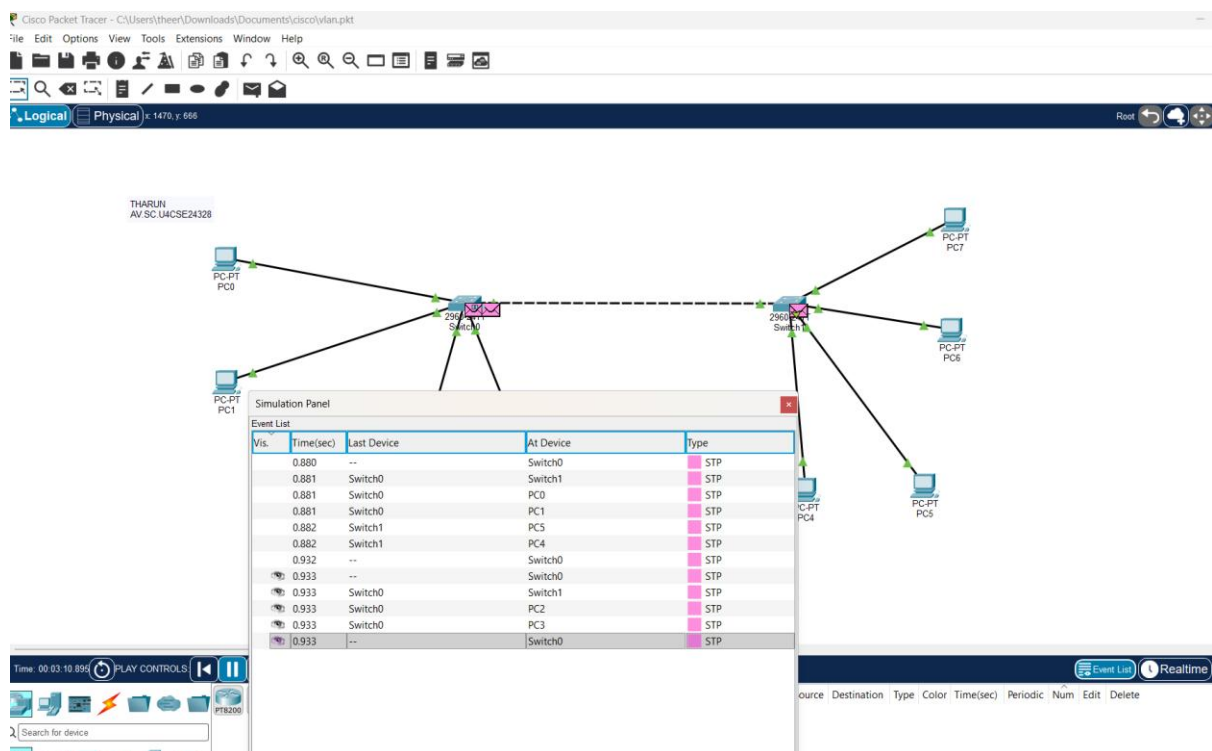
11. Test communication between different VLANs.

From PC0, enter:

ping 192.168.20.1

The ping should fail / show Request Timed Out because PC0 belongs to VLAN 10 and PC2 belongs to VLAN 20, and no router/inter-VLAN routing is configured.

OUTPUT:



Failed Case:

If we try to transfer the packet from one group(CSE) to another group(CAI), the packet will not transfer.

LAB – 7

Aim:

Configure and Enable HTTP and HTTPS Services on a Server in Cisco Packet Tracer and Test Web Connection Between PC and the Server.

Apparatus:

- Cisco Packet Tracer
- 1 × Server
- 2 × PC
- Copper Straight-Through cable
- IP addresses
- Web Browser

Procedure:

1. Open Cisco Packet Tracer and create a new workspace.
2. Place 1 Server and 1 PC in the workspace.
3. Connect the PC FastEthernet0 to the Server FastEthernet0 using a Copper Straight-Through cable.
4. Configure the Server IP address:
 - Go to Server → Desktop → IP Configuration
 - IP Address: 192.168.1.2
 - Subnet Mask: 255.255.255.0

5. Configure the PC IP address:

PC0:

PC0 → Desktop → IP Configuration → Static → IP Address: 192.168.1.3 → Subnet Mask: 255.255.255.0 → Default Gateway: (leave empty) → Close window

PC1:

PC1 → Desktop → IP Configuration → Static → IP Address: 192.168.1.4 → Subnet Mask: 255.255.255.0 → Default Gateway: (leave empty) → Close window

6. Enable HTTP and HTTPS services:
 - Click Server → Services → HTTP.

- Set HTTP: On.
 - Set HTTPS: On.
7. To customize the webpage, edit the index.html file and enter a message such as:
Welcome to my Server
 8. Test basic connectivity from the PC:
 - Go to PC0 → Desktop → Command Prompt.
 - Enter:

ping 192.168.1.2

- Verify that replies are received.
9. Test the HTTP service:
 - Go to PC0 → Desktop → Web Browser.
 - Enter:

http://192.168.1.2

- Press Enter.
- The server webpage should be displayed.

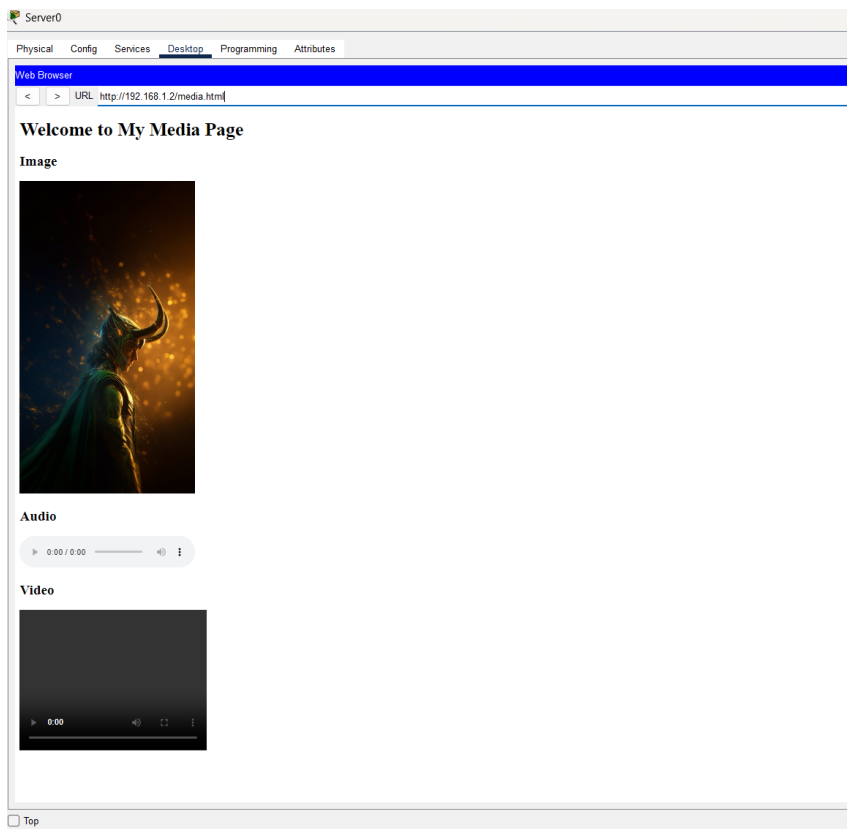
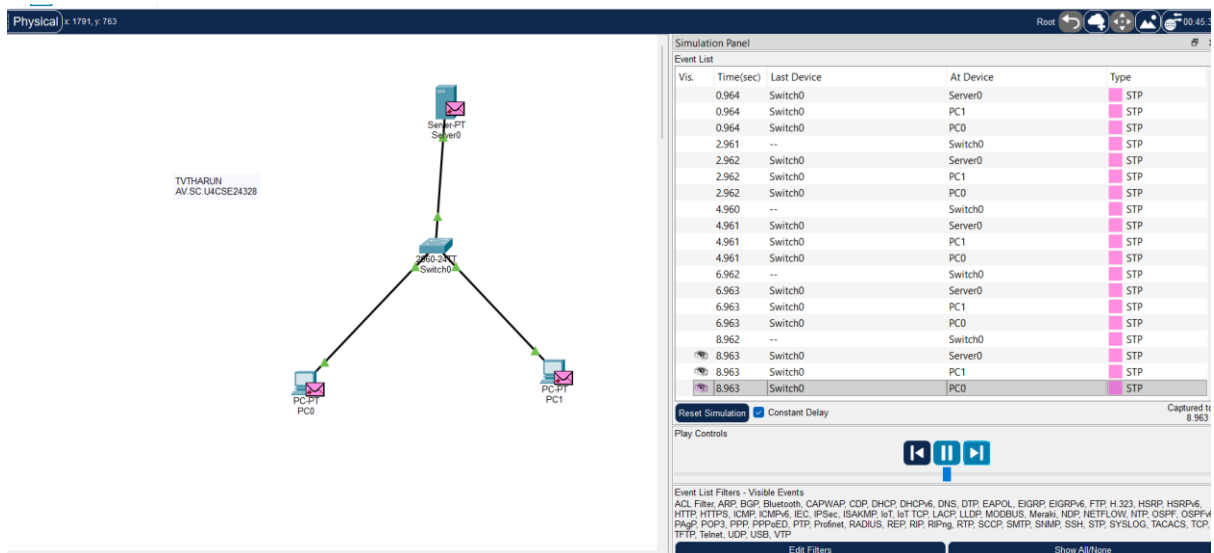
10. Test the HTTPS service:

- In the Web Browser, enter:

https://192.168.1.2

- Press Enter.
- The webpage should load successfully.

OUTPUT:



Failed Case:

1. Wrong URL: Entering an incorrect URL such as `http://192.168.1.5`.
2. Server Down: The server device is powered off or the interface is inactive.
3. Connectivity Failure: ping 192.168.1.2 results in Request Timed Out.
4. Webpage Not Found: The browser displays an error when `index.html` is unavailable or incorrectly configured.

LAB – 8

Aim:

To set up FTP service on the server and do file upload and download between the PCs and the server using FTP commands.

Apparatus:

- Cisco Packet Tracer
- 1 × Server
- 1 × Switch
- 2 × PCs
- Copper Straight-Through cables
- FTP service
- Command Prompt / FTP client

Procedure:

1. Open Cisco Packet Tracer and create a new workspace.
2. Place:
 - 1 Server
 - 1 Switch
 - 2 PCs – PC1 and PC2
3. Connect the Server, PC1, and PC2 to the switch using Copper Straight-Through cables.
4. Configure the IP addresses:

Server:

Server → Desktop → IP Configuration → Static → IP Address: 192.168.1.2 → Subnet Mask: 255.255.255.0 → Default Gateway: (leave empty) → Close window

PC1:

PC1 → Desktop → IP Configuration → Static → IP Address: 192.168.1.3 → Subnet Mask: 255.255.255.0 → Default Gateway: (leave empty) → Close window

PC2:

PC2 → Desktop → IP Configuration → Static → IP Address: 192.168.1.4 → Subnet Mask: 255.255.255.0 → Default Gateway: (leave empty) → Close window

5. Enable the FTP service:

- Click Server → Services → FTP.
- Set FTP Service: On.

6. Create an FTP user:

- Username: tharun
- Password: 1234
- Enable the following permissions:
 - Read
 - Write
 - Delete
 - Rename
 - List
- Click Add and then Save.

7. Test connectivity from PC1:

- Go to PC1 → Desktop → Command Prompt.
- Enter:

```
ping 192.168.1.2
```

10. Start an FTP connection from PC1:

```
ftp 192.168.1.2
```

11. Enter the FTP credentials:

```
Username: tharun
```

```
Password: 1234
```

12. Create a file such as test.txt on PC1 using Text Editor.

13. Upload the file to the server:

```
put test.txt
```

14. Verify that the file has been uploaded:

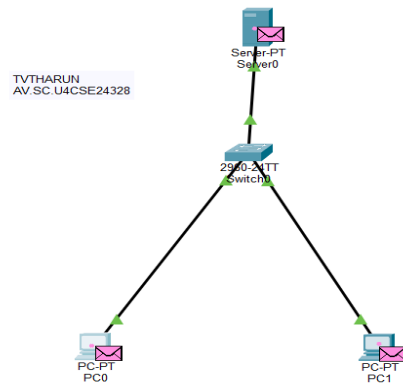
```
dir
```

15. To download a file from the server to PC1, use:

```
get test.txt
```

16. The file will be downloaded to the PC.

OUTPUT:



```

Cisco Packet Tracer PC Command Line 1.0
C:\>ftp 192.168.1.2
Trying to connect...192.168.1.2
Connected to 192.168.1.2
220- Welcome to PT Ftp server
Username:tharun
331- Username ok, need password
Password:****
230- Logged in
(passive mode On)
ftp>
  
```

Simulation Panel				
Event List				
Vis.	Time(sec)	Last Device	At Device	Type
	0.875	--	Switch0	STP
	0.876	Switch0	PC1	STP
	0.876	Switch0	PC0	STP
	0.876	Switch0	Server0	STP
	0.876	--	PC0	FTP
	0.877	PC0	Switch0	FTP
	0.878	Switch0	Server0	FTP
	0.878	--	Server0	FTP
	0.879	Server0	Switch0	FTP
	0.880	Switch0	PC0	FTP
	0.880	--	PC0	FTP
	0.881	PC0	Switch0	FTP
	0.882	Switch0	Server0	FTP
	0.882	--	Server0	FTP
	0.883	Server0	Switch0	FTP
	0.884	Switch0	PC0	FTP
	0.949	--	PC0	TCP
	0.950	PC0	Switch0	TCP

Reset Simulation ☒ Constant Delay Captured to: 0.950 s

Play Controls

Event List Filters - Visible Events
 ACL Filter, ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6,

Failed Case:

1. FTP service OFF: ftp 192.168.1.2 connection fails.
2. Wrong username/password: Login is rejected.
3. Write permission disabled: put test.txt fails.
4. Read permission disabled: get test.txt fails.
5. You can't see the server files because the list permission is disabled for that directory.
6. If a file is not available, you can't upload or download it. This means that if you try to put or get a file that doesn't exist, it will fail.

LAB – 9

Aim:

How to Configure the DNS Service on a Server in Cisco Packet Tracer & To link domain name to IP address to test network connectivity.

Apparatus:

- Cisco Packet Tracer
- 1 × Server
- 1 × Switch
- 2 × PCs
- Copper Straight-Through cables
- DNS service
- HTTP service
- Command Prompt and Web Browser

Procedure:

1. Open Cisco Packet Tracer and create a new workspace.
2. Place:
 - 1 Server
 - 1 Switch
 - 2 PCs – PC1 and PC2
3. Connect the Server, PC1, and PC2 to the switch using Copper Straight-Through cables.
4. Configure the IP addresses:

Server:

Server → Desktop → IP Configuration → Static → IP Address: 192.168.1.2 → Subnet Mask: 255.255.255.0 → Default Gateway: (leave empty) → DNS Server: 192.168.1.2 → Close window

PC1:

PC1 → Desktop → IP Configuration → Static → IP Address: 192.168.1.3 → Subnet

Mask: 255.255.255.0 → Default Gateway: (leave empty) → DNS Server: 192.168.1.2
→ Close window

PC2:

PC2 → Desktop → IP Configuration → Static → IP Address: 192.168.1.4 → Subnet
Mask: 255.255.255.0 → Default Gateway: (leave empty) → DNS Server: 192.168.1.2
→ Close window

5. Enable the HTTP service:

- Go to Server → Services → HTTP.
- Set HTTP: On.

6. Configure the DNS service:

- Go to Server → Services → DNS.
- Set DNS Service: On.

7. Add a DNS record:

- Name: www.mycompany.com
- Type: A Record
- Address: 192.168.1.2
- Click Add and Save.

8. Test DNS using PC1:

- Go to PC1 → Desktop → Command Prompt.
- Enter:

ping www.mycompany.com

9. The domain name should resolve to:

192.168.1.2

and the ping should receive replies.

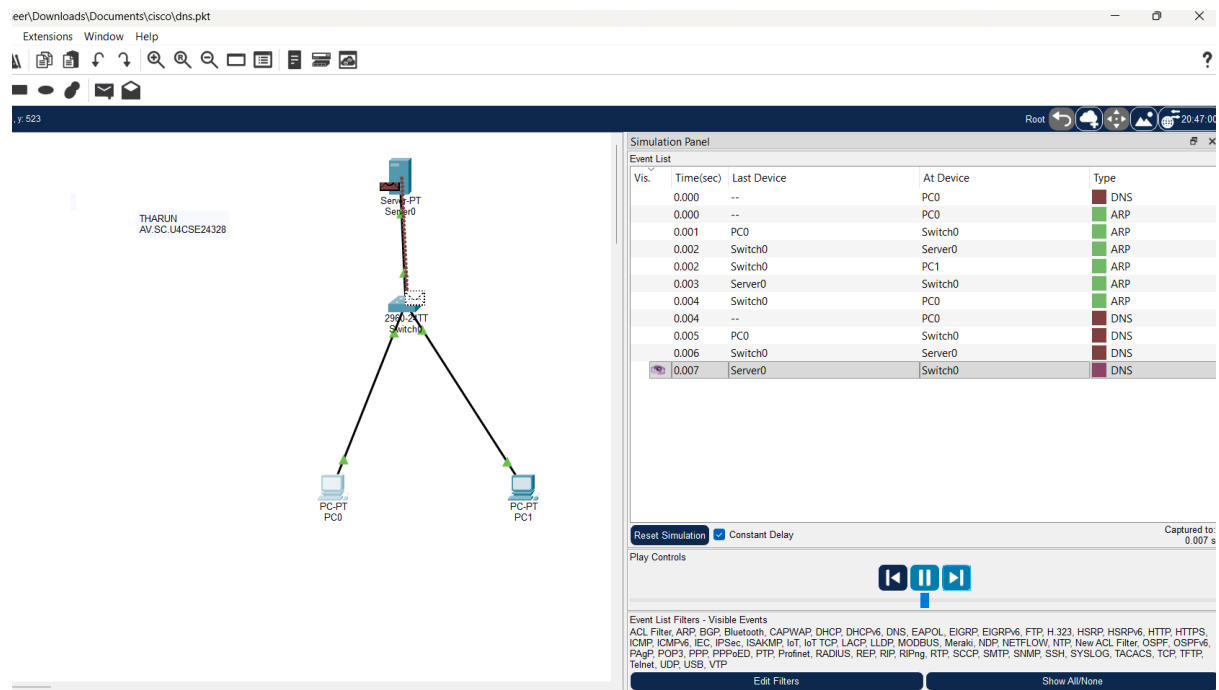
10. Test DNS through the web browser:

- Go to PC1 → Desktop → Web Browser.
- Enter:

http://www.mycompany.com

11. Press Enter. The webpage hosted on the server should load successfully.

OUTPUT:



Failed Case:

1. DNS Service OFF: Domain name will not be resolved.
2. Wrong DNS Server address: PC cannot find the DNS server.
3. Wrong DNS record: Domain points to wrong IP.
4. Wrong domain name: ping or browser request failed.
5. HTTP Service OFF: DNS will resolve correctly but webpage will not load.
6. Wrong IP address: PC can't communicate with server.

LAB – 10

Aim:

To study and compare TCP and UDP protocols in Cisco Packet Tracer, by observing DNS and HTTP traffic and analyzing TCP's 3-way handshake, sequence number, and acknowledgment number.

Apparatus:

- Cisco Packet Tracer
- 1 × Server
- 1 × Switch
- 2 × PCs
- Copper Straight-Through cables
- HTTP and DNS services
- Simulation Mode
- Command Prompt and Web Browser

Procedure:

1. Open Cisco Packet Tracer and create a new workspace.
2. Place:
 - 1 Server
 - 1 Switch
 - 2 PCs – PC1 and PC2
3. Connect the Server, PC1, and PC2 to the switch using Copper Straight-Through cables.
4. Configure the IP addresses:

Server:

Server → Desktop → IP Configuration → Static → IP Address: 192.168.1.2 → Subnet Mask: 255.255.255.0 → Default Gateway: (leave empty) → DNS Server: 192.168.1.2 → Close window

PC1:

PC1 → Desktop → IP Configuration → Static → IP Address: 192.168.1.3 → Subnet Mask: 255.255.255.0 → Default Gateway: (leave empty) → DNS Server: 192.168.1.2 → Close window

PC2:

PC2 → Desktop → IP Configuration → Static → IP Address: 192.168.1.4 → Subnet Mask: 255.255.255.0 → Default Gateway: (leave empty) → DNS Server: 192.168.1.2 → Close window

5. Enable the required services on the Server:

- Go to Server → Services → HTTP and turn HTTP: On.
- Go to Server → Services → DNS and turn DNS: On.

6. Add a DNS record:

- Name: www.test.com
- Type: A Record
- Address: 192.168.1.2
- Click Add and Save.

UDP Traffic – DNS

7. Switch to Simulation Mode.

8. Click Edit Filters and disable:

- STP
- CDP
- DTP

9. Make sure the following are enabled:

- DNS
- HTTP
- TCP
- UDP

10. On PC1 → Desktop → Command Prompt, enter:

ping www.test.com

11. In the Simulation Panel, locate the DNS packet.

12. Double-click the DNS packet to open PDU Information.

13. Expand Layer 4 and observe that DNS uses UDP.

TCP Traffic – HTTP

14. On PC1, open Desktop → Web Browser.

15. Enter:

<http://www.test.com>

16. In the Simulation Panel, locate the TCP packets generated by the HTTP connection.

17. Double-click the first TCP packet and open PDU Information.

18. Expand Layer 4 and observe TCP.

19. Observe the:

- Sequence Number
- Acknowledgment Number

20. Click Play in the Simulation Panel and observe the TCP 3-Way Handshake:

SYN → SYN-ACK → ACK

OUTPUT:

The screenshot shows a network simulation interface. On the left, a topology diagram displays a central switch labeled '2950-24TT Switch0' connected to three devices: 'Server0' (top), 'PC-PT PC0' (bottom left), and 'PC-PT PC1' (bottom right). A text box above the switch reads 'THARUN AV:SC.U4CSE24328'. On the right, the 'Simulation Panel' is open, showing an 'Event List' table. The table has columns for 'Vis.', 'Time(sec)', 'Last Device', 'At Device', and 'Type'. The events listed are DNS and ARP packets. The last event at 0.007 seconds is a DNS packet from Server0 to Switch0. Below the table, there are controls for 'Reset Simulation', 'Constant Delay', and 'Captured to: 0.007 s'. At the bottom, there are 'Event List Filters - Visible Events' and buttons for 'Edit Filters' and 'Show All/None'.

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC0	DNS
	0.000	--	PC0	ARP
	0.001	PC0	Switch0	ARP
	0.002	Switch0	Server0	ARP
	0.002	Switch0	PC1	ARP
	0.003	Server0	Switch0	ARP
	0.004	Switch0	PC0	ARP
	0.004	--	PC0	DNS
	0.005	PC0	Switch0	DNS
	0.006	Switch0	Server0	DNS
	0.007	Server0	Switch0	DNS

Case that failed:

1. DNS Service is OFF, www.test.com cannot be resolved.
2. Wrong DNS record: Domain resolves wrong or not at all.
3. HTTP Service OFF: DNS is OK but webpage cannot be loaded.
4. Wrong IP/DNS configuration: No communication
5. Wrong domain name. DNS query fails
6. TCP filter disabled TCP packets will not be displayed in Simulation Mode.